

VARC Annotations for Eight Representative Case Studies

Demo 1: Coherence Violation

Positive sample

observation: A woman wearing sunglasses, a grey hoodie, a black vest, and leggings walks down a series of wooden steps on a dirt trail surrounded by dry vegetation and trees. She carries a small, light-colored dog inside a bright blue sling bag strapped across her chest. As she descends the path, she moves steadily towards the camera, bringing the dog in the carrier into close view before passing the lens.

causal attribution: The sequence of events perfectly adheres to the linearity of time and the micro-temporal continuity of physical actions. The continuous movement of the woman descending the steps, gradually coming closer to the camera, and eventually passing the lens unfolds in a natural, irreversible chronological sequence without spatial or temporal interruptions.

verdict: The video is real and perfectly conforms to real-world temporal logic.

Negative sample

observation: A woman wearing sunglasses, a grey hoodie, a black vest, and leggings is seen in close view, carrying a small, light-colored dog in a bright blue sling bag while moving towards the camera on a dirt trail. Suddenly, her position resets back to the top of a series of wooden steps surrounded by dry vegetation. An instant later, she is seen passing the camera lens, before abruptly jumping backward again to a mid-descent state walking down the path.

causal attribution: This disjointed jumping of states violates the fundamental linearity of time and the macro-causality of motion, explicitly constituting a Coherence Violation. The continuous progression of the woman's walking action is broken at a micro-temporal level because the chronological order of frames is scrambled. This results in the subject abruptly teleporting between different stages of a single continuous descent (start, end, middle) without any fluid or logical transition.

verdict: The video is forged and contains severe temporal and coherence fallacies.

Demo 2: Causal Reversal

Positive sample

observation: On a wooden surface, two black and blue clamps grip the opposing edges of a green rectangular sheet. The clamps twist and pull outwards. Consequently, the green sheet warps and buckles in the center, subsequently tearing in half and separating into two distinct pieces that are pulled apart by the clamps.

causal attribution: This sequence of events perfectly adheres to macroscopic causality, where the physical cause logically and chronologically precedes the effect. The mechanical tension and pulling force applied by the clamps realistically cause the subsequent deformation and structural failure of the material.

verdict: The video is real and perfectly conforms to real-world causal logic.

Negative sample

observation: On a wooden surface, a green rectangular sheet is seen warping and buckling in the

center, subsequently tearing in half and separating into two distinct pieces. Abruptly, the scene cuts to two black and blue clamps gripping the opposing edges of the intact green sheet, twisting and pulling outwards.

causal attribution: This sequence violates the Linearity and Causality of Time. Presenting the macroscopic effect (the sheet warping and tearing) before its physical cause (the clamps gripping and pulling the intact sheet) constitutes a Causal Reversal, contradicting the fundamental arrow of time.

verdict: The video is forged and contains severe temporal and causal fallacies.

Demo 3: Existence Violation

Positive sample

observation: A suspended cylindrical wicker basket descends onto a wooden table, completely covering a static orange tennis ball. After a brief pause where the ball is concealed inside the basket, the basket is lifted vertically, revealing the tennis ball remaining untouched in its original position on the table.

causal attribution: This phenomenon adheres to the fundamental principle of Conservation of Matter and Morphological Stability. The basket merely acts as a temporary visual cover. Without any explicit physical interaction or external force extracting the ball from beneath the basket, the object maintains its continuous existence and spatial position.

verdict: The video is real and perfectly conforms to real-world physical logic.

Negative sample

observation: A suspended cylindrical wicker basket descends onto a wooden table, completely covering a static orange tennis ball. After a brief pause, the basket is lifted vertically, but the tennis ball is no longer present on the table, and the area under the basket is revealed to be empty wood.

causal attribution: This scenario violates the core physical principle of Conservation of Matter and Morphological Stability, specifically demonstrating an Existence Violation (Disappearance After Uncovering). Since the basket simply covered the tennis ball without any mechanism to remove or alter it, the object should remain consistently in its state and location; its spontaneous disappearance into thin air is physically impossible.

verdict: The video is forged and contains physical fallacies.

Demo 4: Identity Violation

Positive sample

observation: A black and white speckled dog with a blue tag on its collar trots towards the camera on a grassy park lawn, with large trees and parked cars visible in the background. As the dog approaches, it lowers its head to sniff the ground while continuing its forward movement.

causal attribution: This event adheres to the physical principle of Conservation of Identity and Intrinsic Properties. The dog consistently maintains its fundamental biological characteristics, form, and species throughout its continuous physical movement, with no unnatural mutations occurring.

verdict: The video is real and perfectly conforms to real-world physical logic.

Negative sample

observation: A black and white speckled dog with a blue tag on its collar trots towards the camera on a grassy park lawn. Mid-stride, as the dog lowers its head to sniff the ground, it seamlessly and instantly morphs into a sheep while continuing the exact same walking trajectory and forward cadence.

causal attribution: This event violates the principle of Conservation of Identity and Intrinsic Properties, specifically demonstrating an Identity Violation via Spontaneous Transformation (Morphology/Species Mutation). The dog's intrinsic properties undergo a sudden, physically impossible mutation into a completely different species without any corresponding causal interaction or realistic transformative physical process.

verdict: The video is forged and contains physical fallacies.

Demo 5: Dynamic Violation

Positive sample

observation: A yellow rubber duck with an orange beak sits completely stationary on a light brown wooden table. The background is a plain, light grey wall. The lighting casts a soft shadow to the right of the duck, and there is no movement from the object or the camera throughout the sequence.

causal attribution: This scenario perfectly adheres to the principles of classical mechanics and the causal consistency of physical interactions. Specifically, it demonstrates Newton's First Law of Motion (inertia); since no external forces are acting upon the rubber duck, it maintains its state of static equilibrium and remains at rest.

verdict: The video is real and perfectly conforms to real-world physical logic.

Negative sample

observation: A yellow rubber duck with an orange beak sits on a light brown wooden table against a plain, light grey wall with a soft shadow cast to its right. Without any visible external contact, wind, or tilting of the table, the rubber duck suddenly begins to slide smoothly across the wooden surface to the right. It accelerates from a standstill while maintaining its exact static pose throughout its traversal across the table.

causal attribution: This phenomenon violates the 'Causal Consistency of Physical Interactions', specifically representing a 'Phantom Force (Causality without Contact)' dynamic violation. According to Newton's laws of motion, an object's state of static equilibrium cannot be broken, nor can it accelerate, unless a visible external force acts upon it. The rubber duck initiating motion and accelerating completely on its own without any visible actor or physical contact defies classical mechanics.

verdict: The video is forged and contains physical fallacies.

Demo 6: Constraint Violation

Positive sample

observation: A campfire constructed of logs and a stone ring burns brightly on a pebble beach. In the background, gentle ocean waves roll towards the shore near a large white cliff face under a cloudy twilight sky.

causal attribution: This scene adheres to the principles of thermodynamics and natural

environmental physics. The fire is fueled by the logs and oxygen in the open air, burning steadily within the safety of the stone ring, while the ocean waves remain at a distance in the background, causing no interference with the combustion process.

verdict: The video is real and perfectly conforms to real-world physical logic.

Negative sample

observation: A campfire constructed of logs and a stone ring is situated on a pebble beach near a large white cliff. Ocean waves surge forward and completely engulf the campfire. Once fully submerged, the bright orange flames continue to burn and flicker rhythmically underwater without being extinguished and without generating any steam.

causal attribution: This sequence violates the 'Causal Consistency of Physical Interactions', specifically exhibiting a 'Thermodynamic & Environmental Disconnect'. A standard wood-fueled campfire requires gaseous oxygen to sustain combustion. When engulfed by water, the rapid loss of heat and oxygen should immediately extinguish the fire and produce steam. The flames continuing to burn steadily underwater blatantly contradicts the thermodynamic interactions and material constraints between liquid water and an open flame.

verdict: The video is forged and contains physical fallacies.

Demo 7: Near Miss

Positive sample

observation: A soccer ball rests on green artificial turf against a dark background. A player wearing white socks, blue shorts, and cleats approaches from the left, swings their right leg, and brings their foot into direct physical contact with the ball. Upon this impact, the ball is launched upward and forward into the darkness, while the player continues running through the frame.

causal attribution: The observed sequence strictly aligns with the physical laws of kinetic energy transfer. Because the player's swinging foot makes direct physical contact with the stationary ball, force is successfully applied to the object. This explicit visual fact dictates the physical outcome, causing the ball to be launched along the path of the applied force.

verdict: The video is a visually authentic recording that perfectly conforms to physical logic.

Negative sample

observation: A soccer ball rests on green artificial turf against a dark background. A player wearing white socks, blue shorts, and cleats approaches from the left and forcefully swings their right leg. The foot travels on a trajectory that passes millimeters beside the soccer ball, maintaining a visible spatial gap without touching it. The soccer ball remains completely stationary and undisturbed on the grass while the player continues running through the frame.

causal attribution: This is a prime example of a 'Near-Miss Interaction', specifically a 'Trajectory Miss'. Although the running and forceful swinging motion strongly triggers a semantic prior of a soccer kick, the visual facts show a strict spatial gap between the foot and the ball in the interaction zone. Because zero physical contact occurs, no kinetic energy is transferred, and the expected causal chain reaction rightfully does not happen. The ball remaining absolutely stationary perfectly aligns with the explicit pixel evidence.

verdict: The video is a visually authentic recording that perfectly conforms to physical logic.

Demo 8: Consequence Arrest

Positive sample

observation: A yellow ceramic mug drops vertically from a blue hanging clip against a white background. It falls and strikes a concrete block resting on a wooden deck. Upon contact, the mug breaks into multiple shards that scatter across the floor.

causal attribution: The physical logic aligns directly with the observed facts of the interaction. The visual evidence shows the mug shattering upon striking the concrete block, indicating that the impact force was sufficient to exceed the structural integrity threshold of the material, resulting in structural failure.

verdict: The video is a visually authentic recording that perfectly conforms to physical logic.

Negative sample

observation: A yellow ceramic mug drops vertically from a blue hanging clip against a white background. It falls and strikes a concrete block resting on a wooden deck. Upon contact, the mug remains completely intact with zero structural damage. It bounces and topples off the concrete block onto the wooden deck as a solid, unbroken unit.

causal attribution: This interaction demonstrates 'Consequence Arrest' specifically due to 'Insufficient Impact.' Although the mug physically strikes the concrete block, the explicit visual facts show it remaining completely intact. This indicates that the impact force or material dynamics failed to cross the critical physical threshold required to cause structural damage. The physical outcome strictly aligns with the observed visual facts, correctly overriding the semantic expectation or script that a ceramic object must shatter against concrete.

verdict: The video is a visually authentic recording that perfectly conforms to physical logic.